



Business Problems – Road Accident Risk Analysis

1. Dataset Exploration

Problem:

Understand the structure, fields, and overall distribution of accident data to enable further risk analysis and decision-making.

2. Top City per State by Accident Count

Problem:

Identify the most accident-prone city in each state to help authorities prioritize high-risk urban areas for safety interventions.

3. State-wise Risk Analysis

Problem:

Evaluate and compare average accident risk across states to determine which regions are more vulnerable and require targeted policies.

4. Accident Count by Cause

Problem:

Analyse the primary causes of accidents to identify key risk factors and support preventive measures such as awareness campaigns and law enforcement.

5. Top High-Risk Cities

Problem:

Identify cities with the highest average accident risk to focus infrastructure improvements and safety regulations in critical zones.

6. Risk by Vehicle Category

Problem:

Assess how accident risk varies with the number of vehicles involved to understand collision severity and improve traffic management strategies.

7. Weekend Urban Accidents

Problem:

Determine accident frequency in urban areas during weekends to evaluate the impact of increased leisure travel and traffic congestion.

8. Peak Hour Urban Accidents

Problem:

Analyze accident occurrences during peak traffic hours in urban areas to optimize traffic control and reduce congestion-related risks.

9. Fatal Accidents by Cause

Problem:

Identify key causes contributing to fatal accidents to implement stricter safety measures and reduce mortality rates.

10. Festival vs Normal Day Risk

Problem:

Compare accident frequency and risk during festival days versus normal days to understand the impact of crowd movement and seasonal travel.

11. Most Dangerous Day of the Week

Problem:

Determine which day of the week has the highest accident frequency to improve resource allocation such as traffic police deployment and emergency services.